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26 April 1988
SUPERSEDING
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15 June 1987

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5847/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).

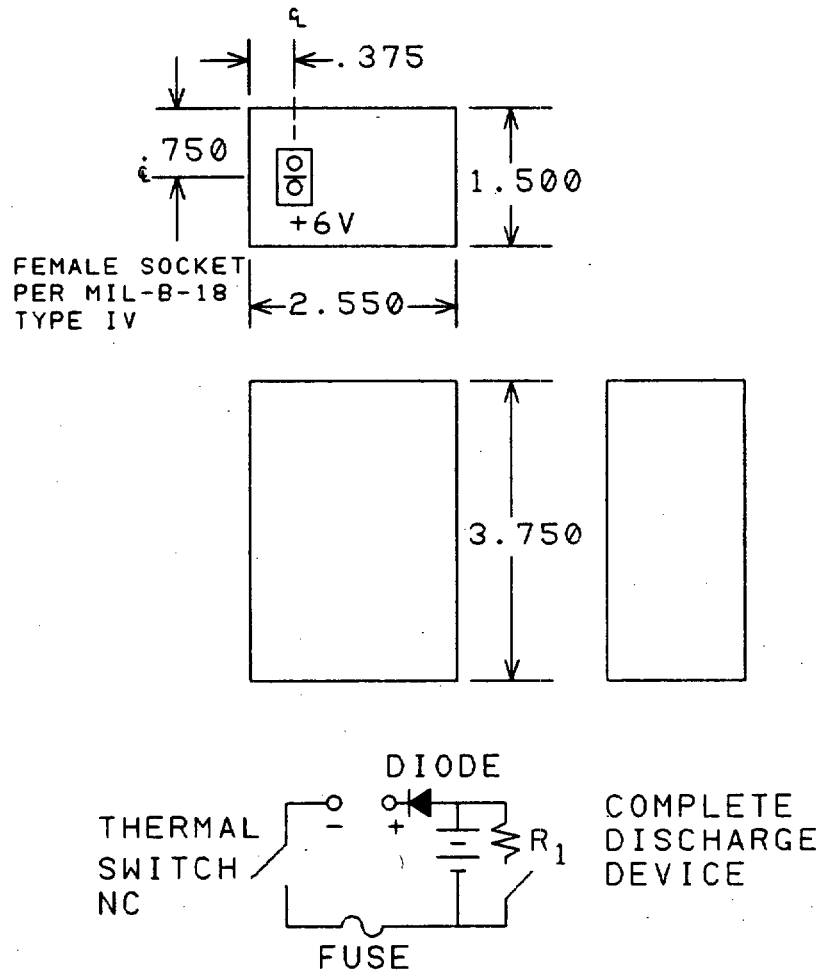


Figure 1. Battery dimensions and wiring schematic.

FSC 6135

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NOTES:

1. All dimensions are in inches.

2. Tolerances are ±.031 inch.

3. The minimum depth of the socket well shall be 5/8 inch.

4. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/8 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug. Pin circle center shall be on the battery center line.

5. Socket shall float 1/32 inch in all directions from the specified pin circle center location in the plane of the socket.

6. R_1 value shall be between 60 ohms and 84 ohms and have a power rating of 1.0 watt.

7. Socket wiring schematic shall be printed or embossed on the battery jacket near the connector on the terminal face of the battery.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Maximum Voltage: 6.10 volts.

Minimum voltage for capacity: 4.00 volts.

The battery shall contain a 2.25 ampere non-replaceable, slow blow fuse wired in series with the cells in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the Basic Specification.

Terminal: Two (2) hole female socket per MIL-B-18, Type IV.

Weight (maximum): 0.53 pound (240 grams)

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Capacity Tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in hours to 4.00 volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
T	2.8
D	2.8
ILR	1000

Initial Voltage delay: During the I, H, HT, T, D, and ILR discharge tests initial voltages below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltage below 3.80 volts cannot exceed a 60.0 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:
Capacity tests:

1. Storage: Details in storage conditions for all capacity tests are specified in the basic specification.(See 4.8.9)

2. Discharge:

   Cells: Discharge at 2.0 amperes constant current at $70\pm5^{o}F$ for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 5.4 hours into voltage reversal.

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Batteries: For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. Their shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 1050 ohms shall be used to cut-off voltage of 4.00 volts or 1000 hours.

Closed circuit voltage test: Load resistance of 5 ohms shall be used. Voltage shall be above 4.00 volts within 5 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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